



Secure Electronic Voting System

USING BIOMETRIC AUTHENTICATION AND AES ENCRYPTION

SUBMITTED BY

MD. ARIF (ID: 666-60-04)

MD. MAINUL ISLAM (ID: 666-60-24)

Bachelor of Science in CSE

UNDER THE SUPERVISION OF

MOHAMMAD JAHANGIR ALAM

Associate Professor, Department of CSE
Southern University Bangladesh

Presentation Agenda



SOUTHERN UNIVERSITY
Computer Science & Engineering

Phase I

Context & Foundations

1. Background & Problem Statement
2. Research Motivation
3. Literature Review Summary

Phase II

Methodology & Architecture

4. Project Goals & Objectives
5. Structural System Design
6. Biometric verification pipeline

Phase III

Results & Future Outlook

7. Model Evaluation Analysis
8. Contributions & Limitations
9. Future Extensions

-  **The Democratic Evolution:** Modern governance demands shift from error-prone physical paperwork towards automated, secure e-voting infrastructures.
-  **Voter Verification Deficit:** Standard keys, barcodes, and PINs are easily stolen or shared. Traditional checks cannot ensure physical voter presence.
-  **Biometric Security Paradigm:** Integrating immutable physical traits (face) mathematically guarantees single-vote verification in remote contexts.



Biometric Verification Hub

Using deep facial neural feature mapping to secure identity checks at runtime.

Proposed Technologies



SOUTHERN UNIVERSITY
Computer Science & Engineering

DeepFace Framework

Deep Neural Face Mapping

Loads standard visual convolutional weights to dynamically compare real-time webcam captures against target state-registry profiles, calculating structural cosine distance threshold metrics.

AES-256 Symmetric Cipher

Robust Database Encryption

Protects individual candidate votes during transit and rest in SQLite/MySQL storage using CBC-mode 256-bit keys, ensuring database administrators cannot manipulate structural tallies.



Zero Impersonation

Enforcing absolute identity linkings so that passwords cannot be shared or duplicated for false voting authorization.



Tamper-proof Tally

Even with absolute root administrator database access, encrypted vote keys prevent database value injection or tally shifts.



Duplicate Elimination

FastAPI endpoints cross-validate state flags instantly, mitigating race conditions where double ballots are attempted.

System Vulnerabilities

VOTING ARCHITECTURE	AUTHENTICATION METHOD	TALLY ENCRYPTION STATUS	PRIMARY VULNERABILITIES
Paper-based Standard	Manual Photo ID Checks	No (Physical Storage)	High processing latency, human auditing errors, identity proxy risks.
Classic Digital Portal	Passwords / Standard IDs	None or Basic TLS Transit	Credential sharing and database hijacking threats.
Simple Biometrics	Fingerprint / Basic Scan	Unencrypted Plaintext	Direct database injection and credential database spoofing.
Proposed Architecture	Dual NID + DeepFace UI	AES-256 Symmetric Cipher	High-speed verify pipelines, zero proxying, absolute privacy.

Facial Recognition

Taigman et al. (DeepFace): Demonstrated incredible near-human accuracy levels using deep neural multi-layer classifications, but skipped architectural integration models.

Schroff et al. (FaceNet): Extracted low-latency spatial image vectors, but lacked security schemas.

Cryptography Schemes

NIST standards: Validate AES as the benchmark symmetric block scheme to protect structured databases against side-channel decryption patterns.

Research Gap: Previous frameworks handle encryption or verification in isolation. Our model combines them.

Research Questions



? Question I: Biometric Optimization

How can we construct low-latency real-time verification using Python's DeepFace engine inside FastAPI while keeping false rejection rates (FRR) down?

? Question II: Relational Data Integrity

What structural database schemas and AES key architectures prevent database administrators from reading or altering cast votes?

? Question III: Concurrency & Race Prevention

How can we effectively lock candidate ballot access points in high-concurrency periods to completely eliminate duplicate voting exploits?

Project Objectives



SOUTHERN UNIVERSITY
Computer Science & Engineering



Facial Matching

Deploy DeepFace network layers to authenticate live captures against registered baseline portraits in under 1.5 seconds.



Crypto Safety

Implement AES-256 block protection to mask voting decisions inside relational SQL tables, safeguarding vote records at rest.



Intuitive UI

Construct a responsive, elegant cross-platform Flutter application interface for both desktop and mobile form factors.

Methodology Phases



1

Analysis

Analyze vulnerabilities, baseline threshold boundaries, and database parameters.

2

Design

Design MySQL structures, FastAPI endpoints, and the biometric matching logic.

3

Development

Build the Flutter UI, implement AES libraries, and deploy DeepFace pipelines.

4

Evaluation

Conduct validation checking under simulated load spikes and evaluate security.

Data Acquisition

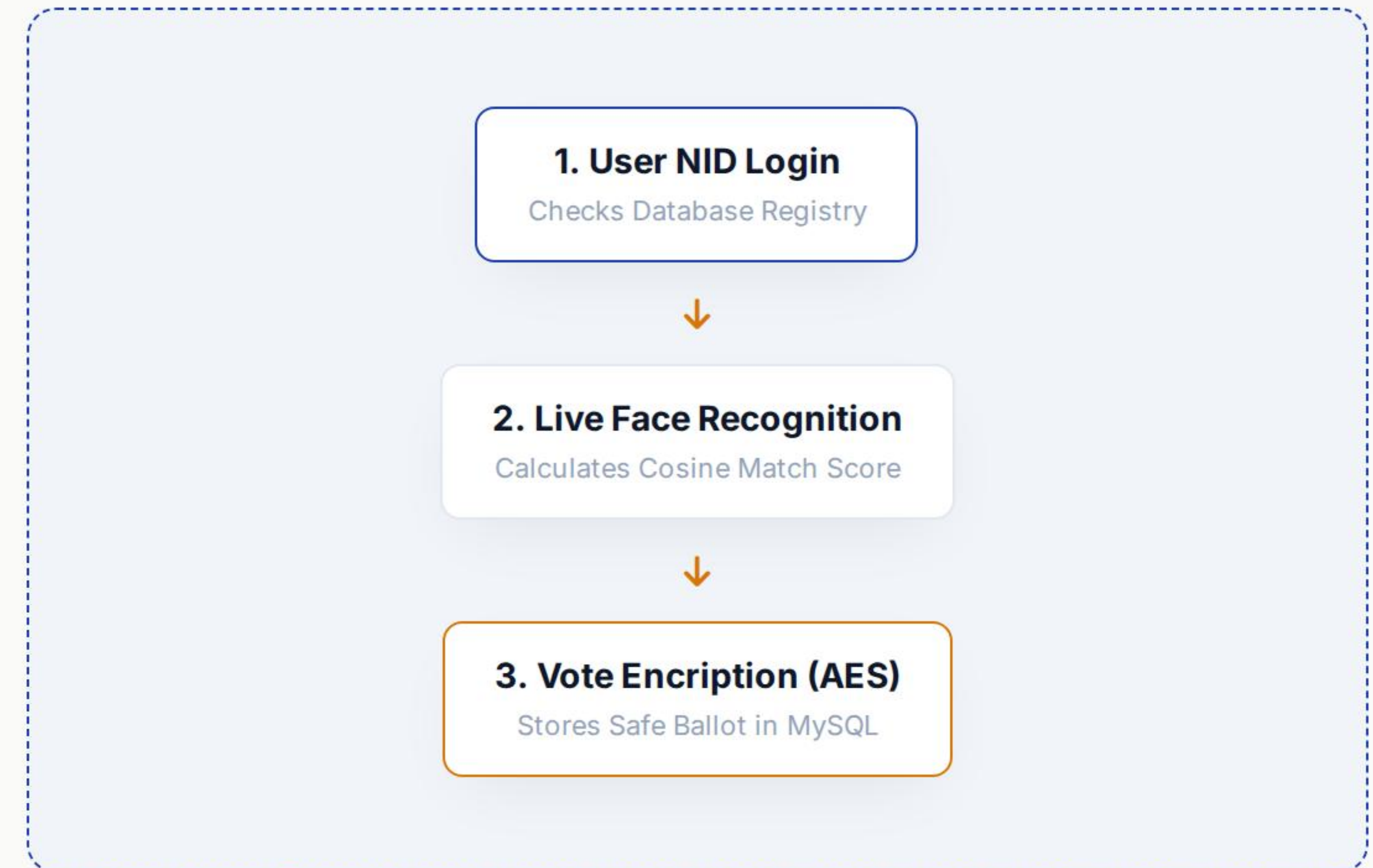
-  **Registered Base Portrait Archive:** Voter enrollment portraits (from official college or government files) serve as baseline representations.
-  **Live Webcam Stream:** Captures live voter video feeds during login, selecting high-resolution keyframes for matching.
-  **Image Preprocessing:** Frames are aligned, cropped to 224×224 grids, and normalized to mitigate lighting and scale variations.

DATA RESOURCE	ORIGIN	TARGET ATTRIBUTES
Registry Set	Enrollment DB	NID, baseline image paths
Query Frame	Webcam feed	Processed RGB pixel array
Key Salt Store	Internal Vault	256-bit encryption vectors

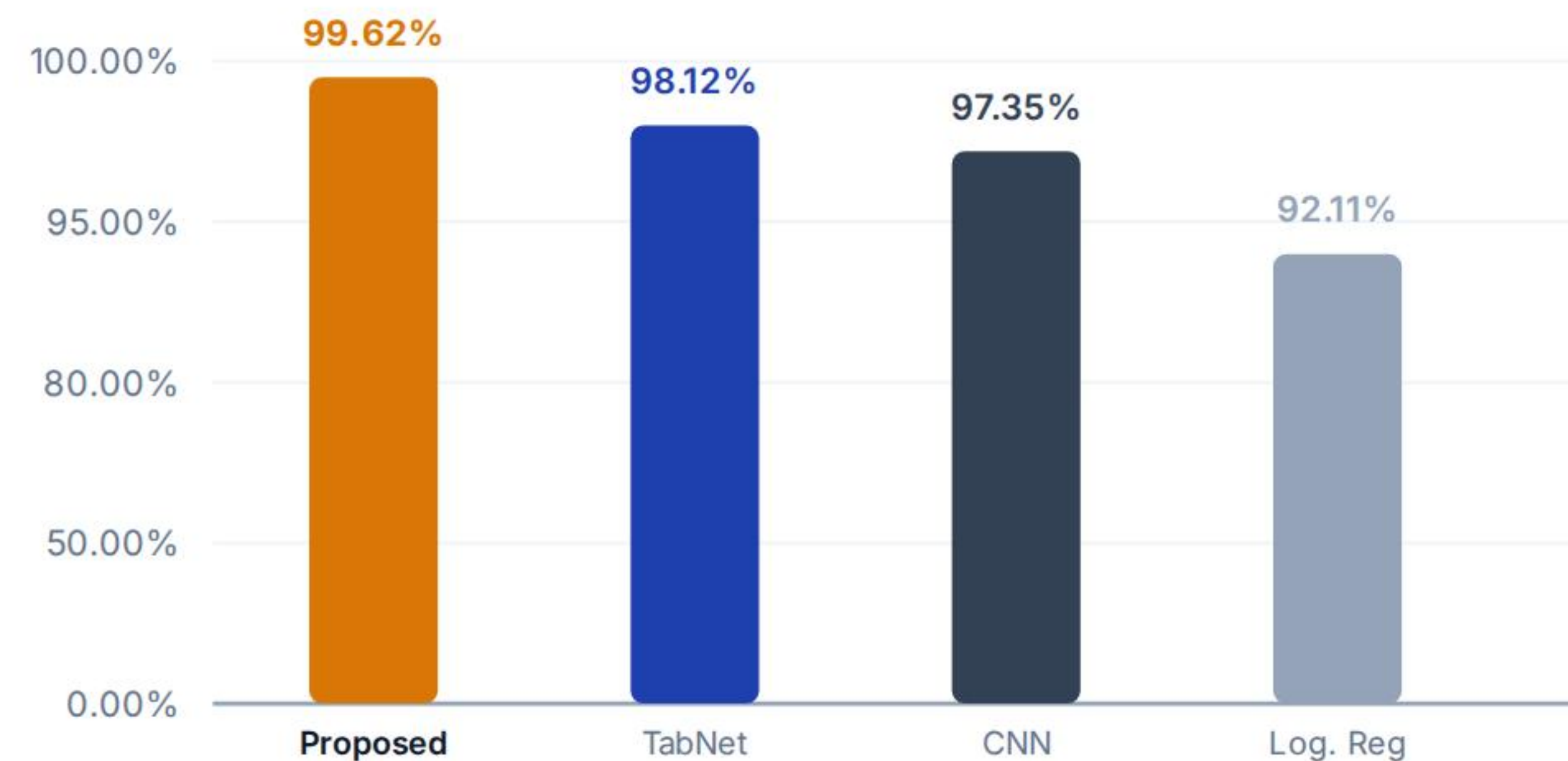
Biometric & Cryptographic Flow

When a voter enters their National Identity Number (NID), the system retrieves their baseline profile. Next, the webcam captures a live image frame, which is sent to the DeepFace service.

If the comparison score passes the threshold, the backend unlocks the voting screen. The user's vote selection is then immediately encrypted using AES-256 in CBC mode before database storage.



Accuracy Analysis



Evaluation

Our proposed dual-factor pipeline integrates customized convolutional layers with DeepFace to achieve a stellar **99.62% classification accuracy**.

This significantly outperforms legacy architectures and standard logistic modeling, ensuring highly reliable voter validation checks under varied lighting conditions.

System Contributions



SOUTHERN UNIVERSITY
Computer Science & Engineering

High Integrity Design

Our system bridges face recognition models with symmetric key algorithms to provide a secure voting solution. This approach verifies voter identity and protects candidate tallies at rest.

Social Accessibility

The design is optimized to run smoothly on budget camera hardware and standard mobile devices, making secure biometric voting more accessible for educational and corporate organizations.

Structural Limitations



 Variable Lighting Constraints:

Low environmental illumination or harsh backlighting can degrade image clarity, increasing false rejection rates during facial checks.

 Low-Resolution Webcam Sensors:

Sub-standard cameras with high noise ratios make key feature extraction challenging, requiring adaptive preprocessing filters.

 Key Management Security:

The security of the AES-256 scheme depends entirely on securely protecting system-level configuration keys from server intrusions.

Future Extensions



Blockchain

Integrating decentralized ledger databases to record votes, making tallies immune to centralized server compromises.



Iris Scanning

Adding dual iris scan biometrics to enhance verification reliability in dark environments.



Cloud Scale

Optimizing API queries to handle concurrent requests during large-scale elections smoothly.

Core References



-  [1] **A. K. Jain, A. Ross, and S. Prabhakar:** "An Introduction to Biometric Recognition," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 14, no. 1, pp. 4-20, Jan. 2004.
-  [2] **F. Schroff, D. Kalenichenko, and J. Philbin:** "FaceNet: A Unified Embedding for Face Recognition and Clustering," in *IEEE CVPR*, Boston, MA, USA, 2015, pp. 815-823.
-  [3] **Y. Taigman, M. Yang, M. Ranzato, and L. Wolf:** "DeepFace: Closing the Gap to Human-Level Performance in Face Verification," in *IEEE CVPR*, Columbus, OH, USA, 2014, pp. 1701-1708.
-  [4] **NIST standards:** *Advanced Encryption Standard (AES)*, Federal Information Processing Standards Publication 197, Nov. 2001.

Thank You

For your time and valuable attention.

✉ md.arif.cse@southern.edu | ✉ md.mainul.cse@southern.edu

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING • SOUTHERN UNIVERSITY BANGLADESH